

6 (a) extension is proportional to force (for small extensions) B1 [1]

(b) (i) point beyond which (the spring) does not return to its original length when the load is removed B1 [1]

(ii) gradient of graph = 80 N m^{-1} A1 [1]

(iii) work done is area under graph / $\frac{1}{2} Fx$ / $\frac{1}{2} kx^2$
 $= 0.5 \times 6.4 \times 0.08 = 0.256$ (allow 0.26) J C1
A1 [2]

(c) (i) extension = $0.08 + 0.04 = 0.12 \text{ m}$ A1 [1]

(ii) spring constant = $6.4 / 0.12 = 53.3 \text{ N m}^{-1}$ A1 [1]

7 (a) nuclei with the same number of protons

B1